

# Technical Insights: Workload Identity Management — Resolve IAM Technical Debt

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**Bad IAM practices is  
compounding interest's evil twin,  
multiplying problems instead of profits.**

**Forge machine  
IAM teams, define  
processes and sharpen  
your tooling support.**



# Key Issues

1. How should I forge machine IAM teams?
2. How should I define my workload IAM processes?
3. How do I sharpen my tooling support?

# We Failed by Keeping Our Old Siloes

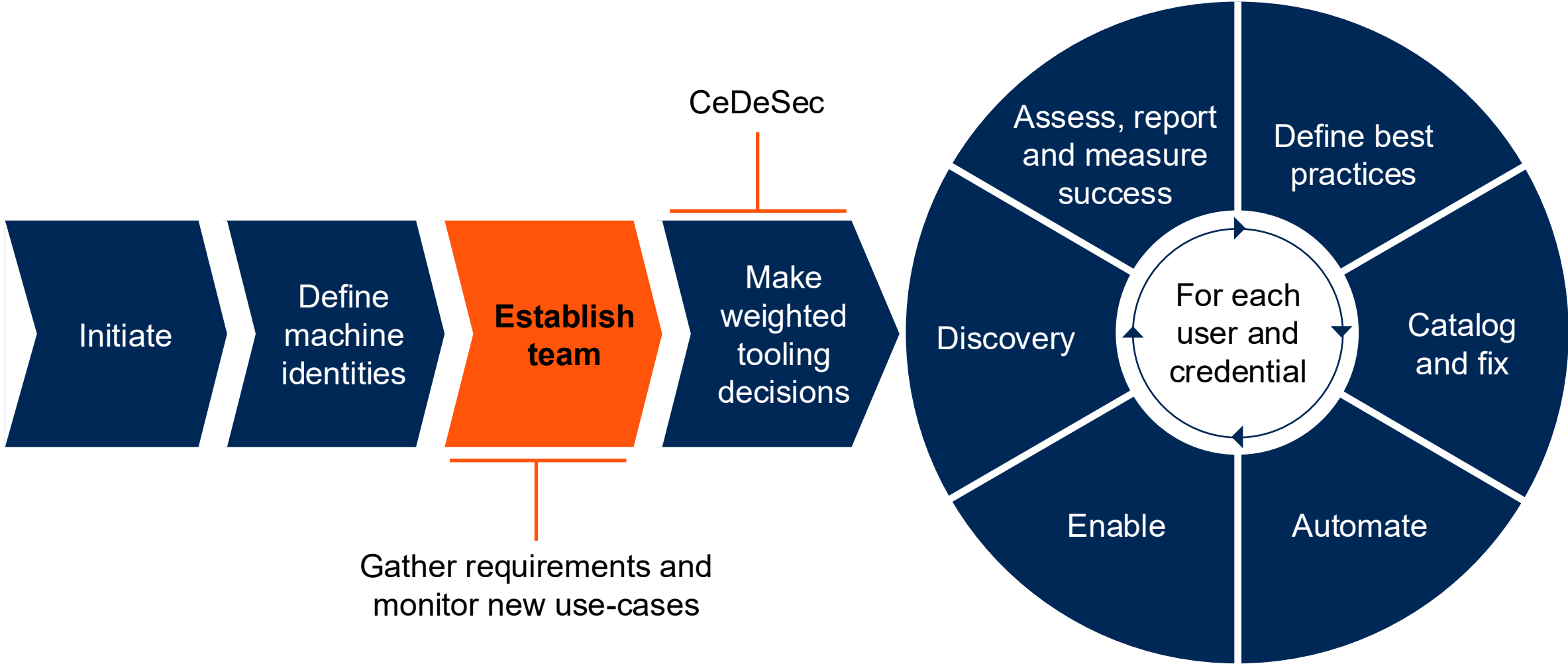


# How to Start the Program?

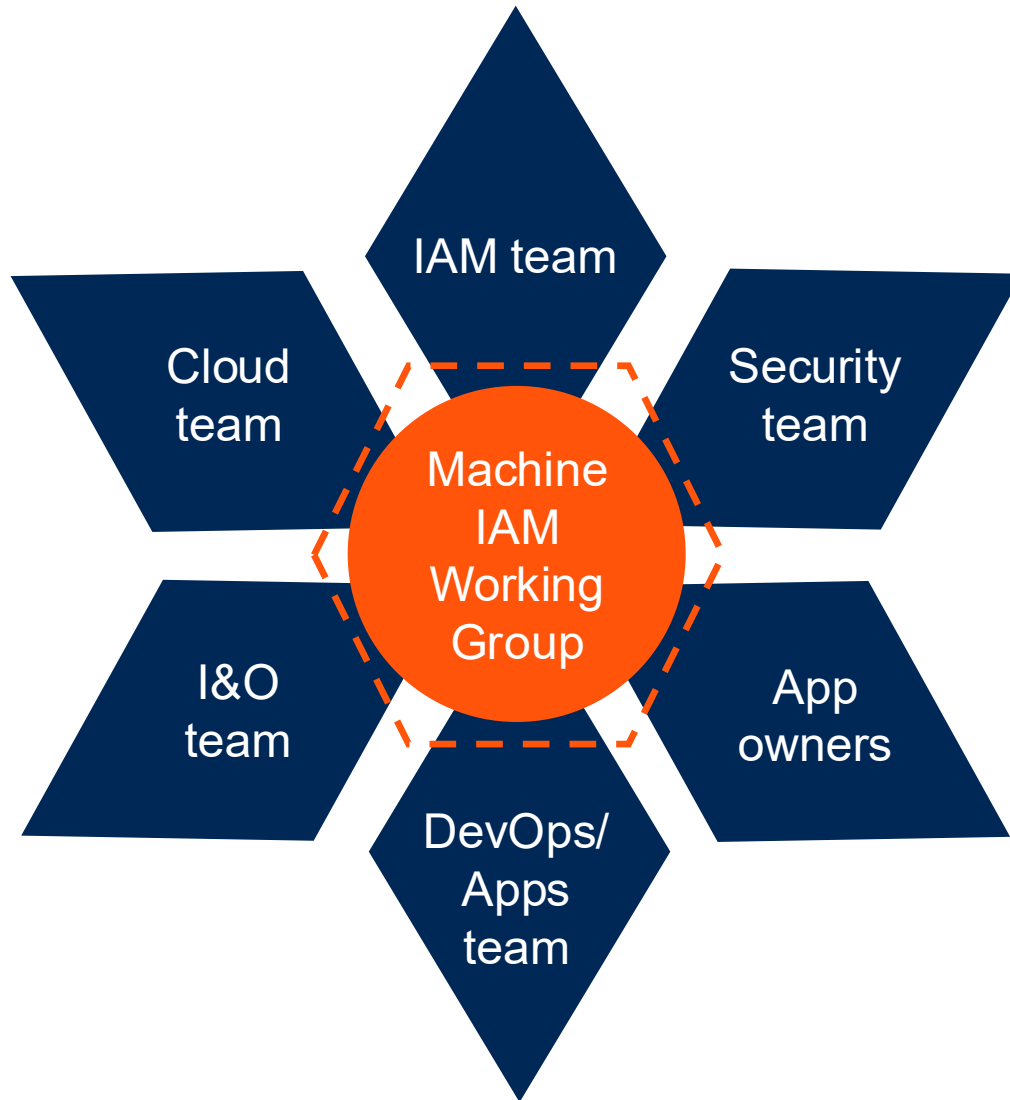
Establish principles, processes and a platform that support the net new

Resolve IAM technical debt

# Resolve IAM Technical Debt



# The Machine IAM Working Group



## Machine identity working group mission:

- Provide leadership.
- Conduct research and gather requirements.
- Lay out strategies.
- Define ownership.
- Deliver targeted best practices.
- Set expectations.
- Be a single point of contact.

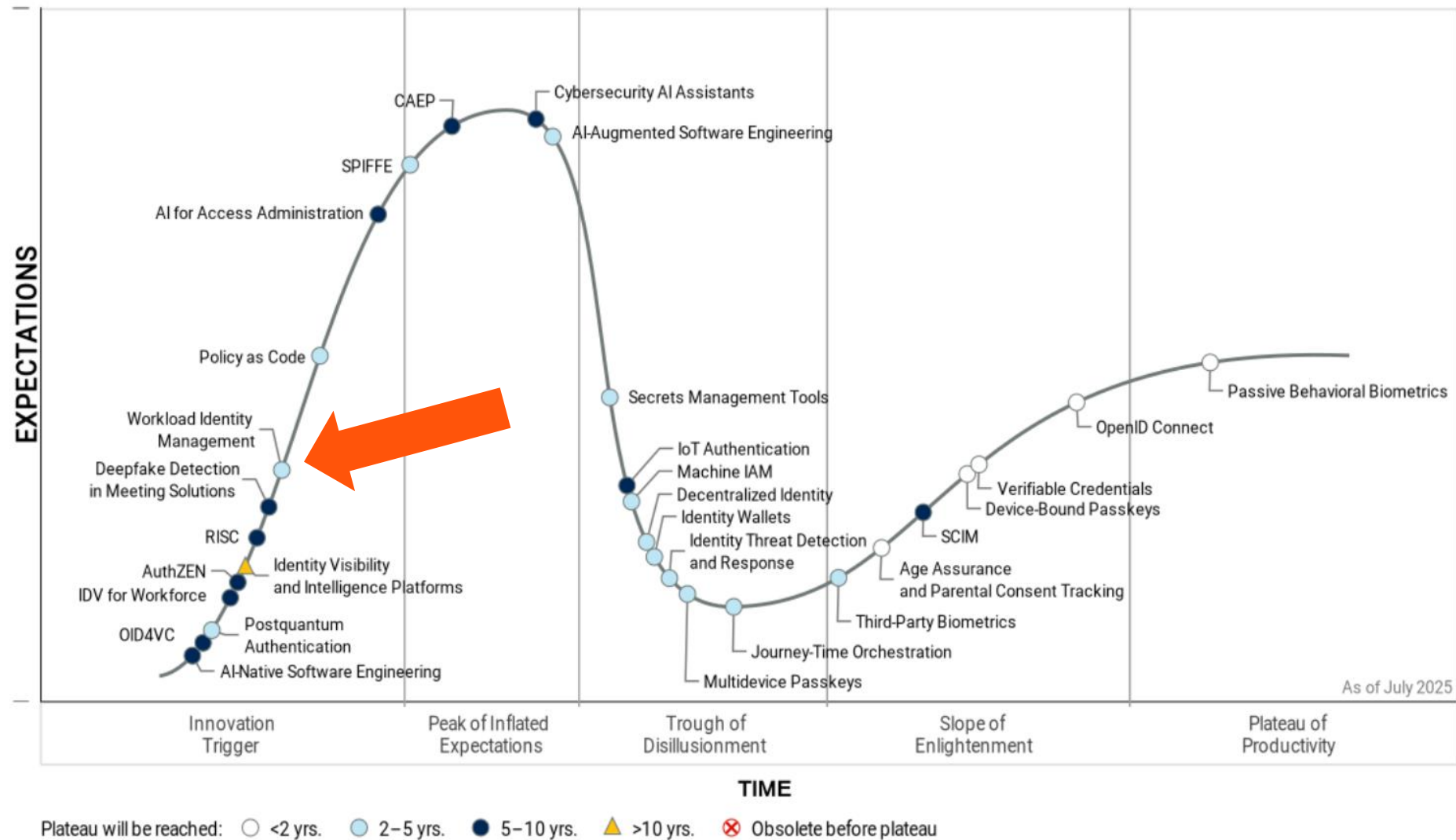
## **Key Issue Take-Away:**

Form cross-functional teams to fix technical debt from poor workload IAM decisions.

# Key Issues

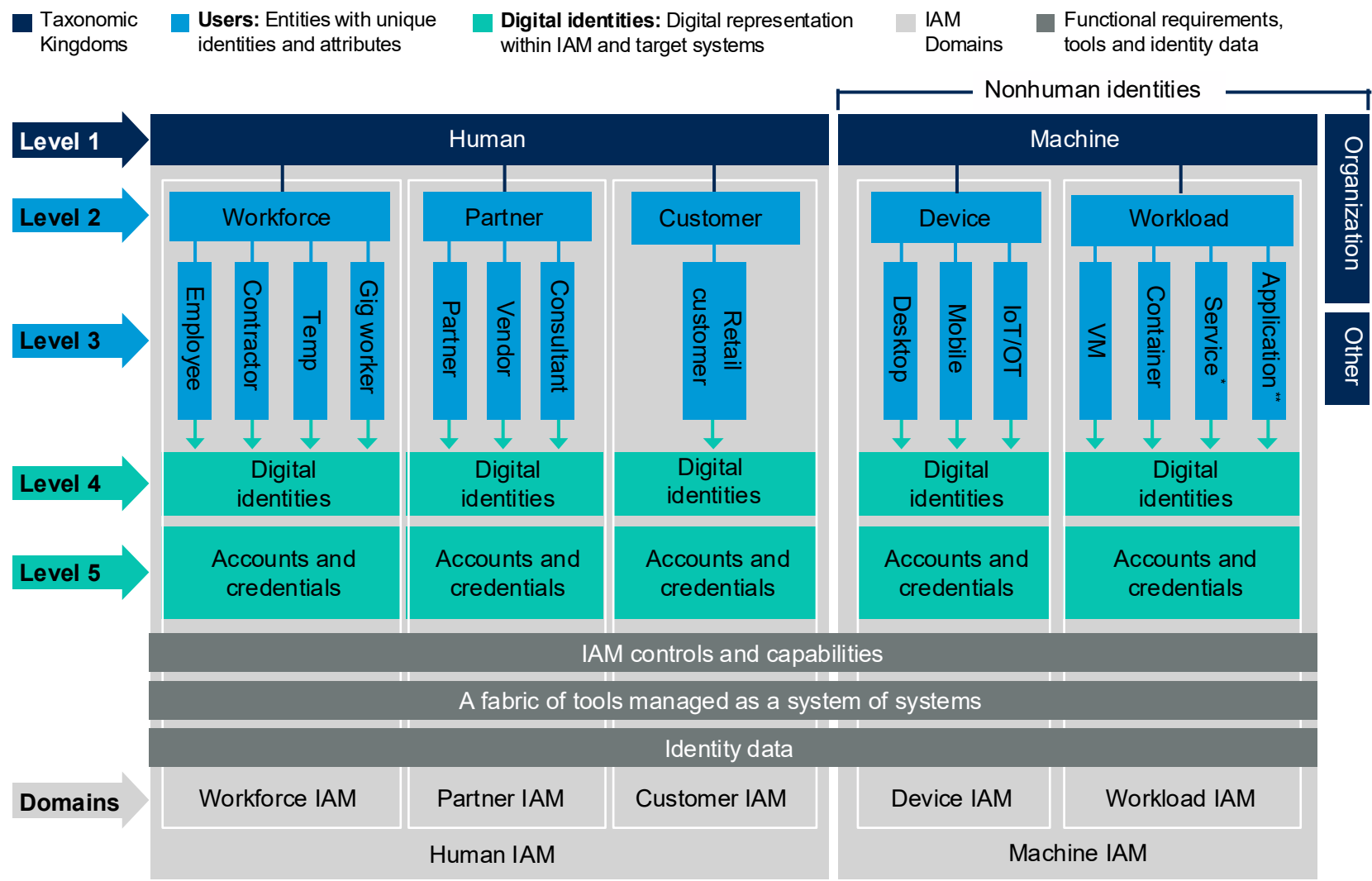
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# Hype Cycle for Digital Identity, 2025



Source: [Hype Cycle for Digital Identity, 2025](#), 14 July 2025

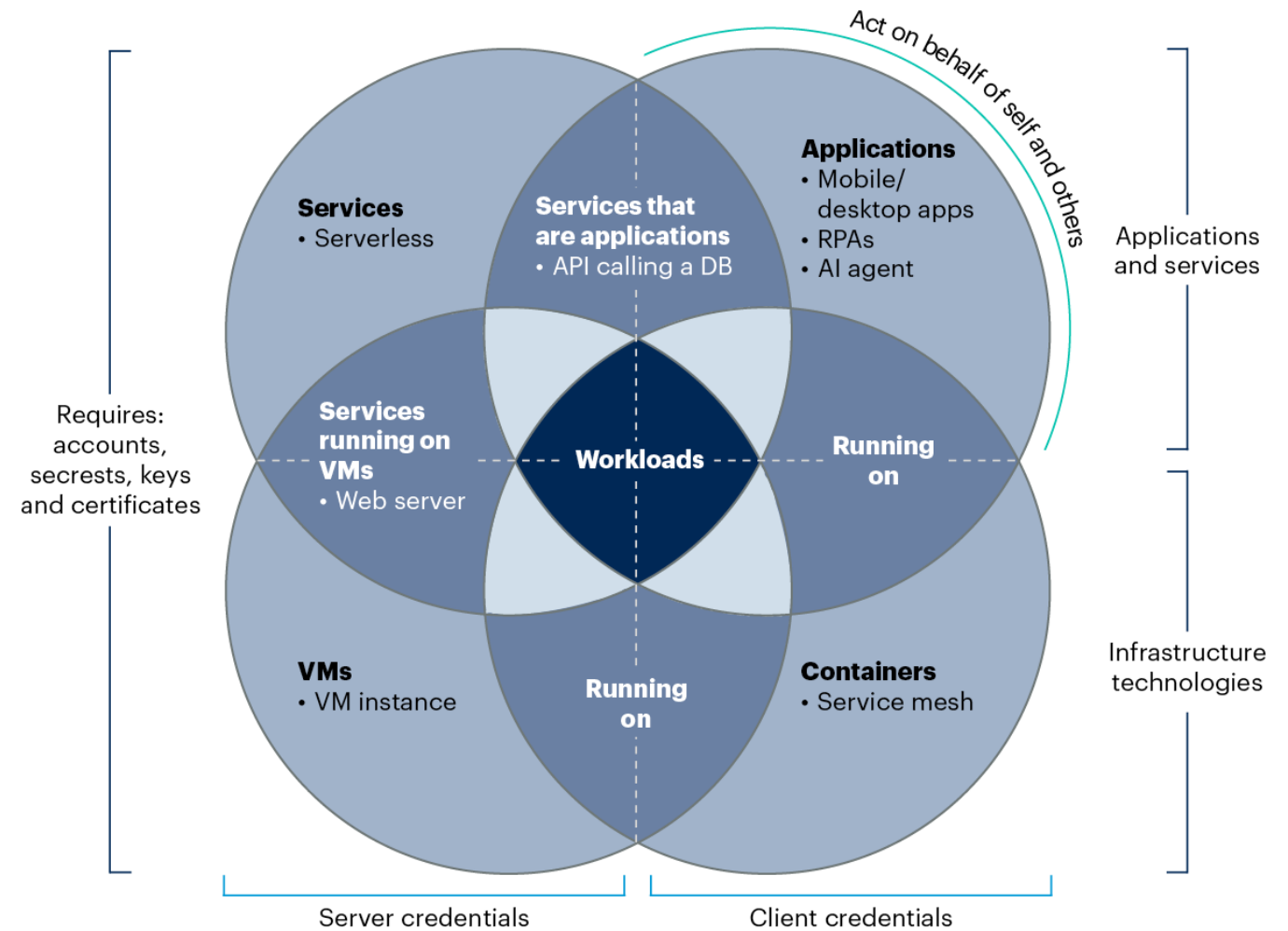
# The Gartner IAM Taxonomy



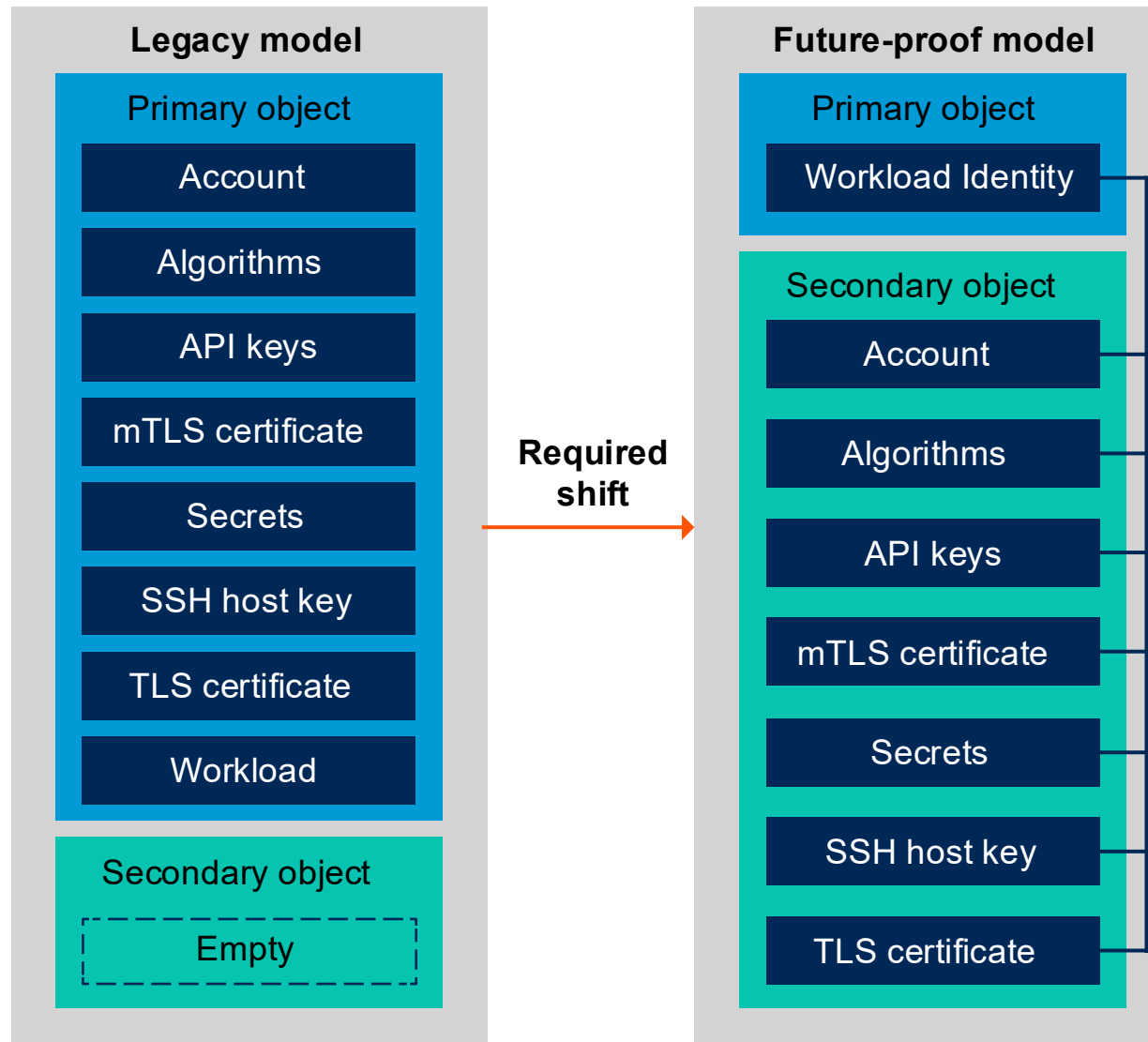
\* = web, REST, API, RPC, webhook  
\*\* = web, native, script, RPA, AI agent

# Workloads

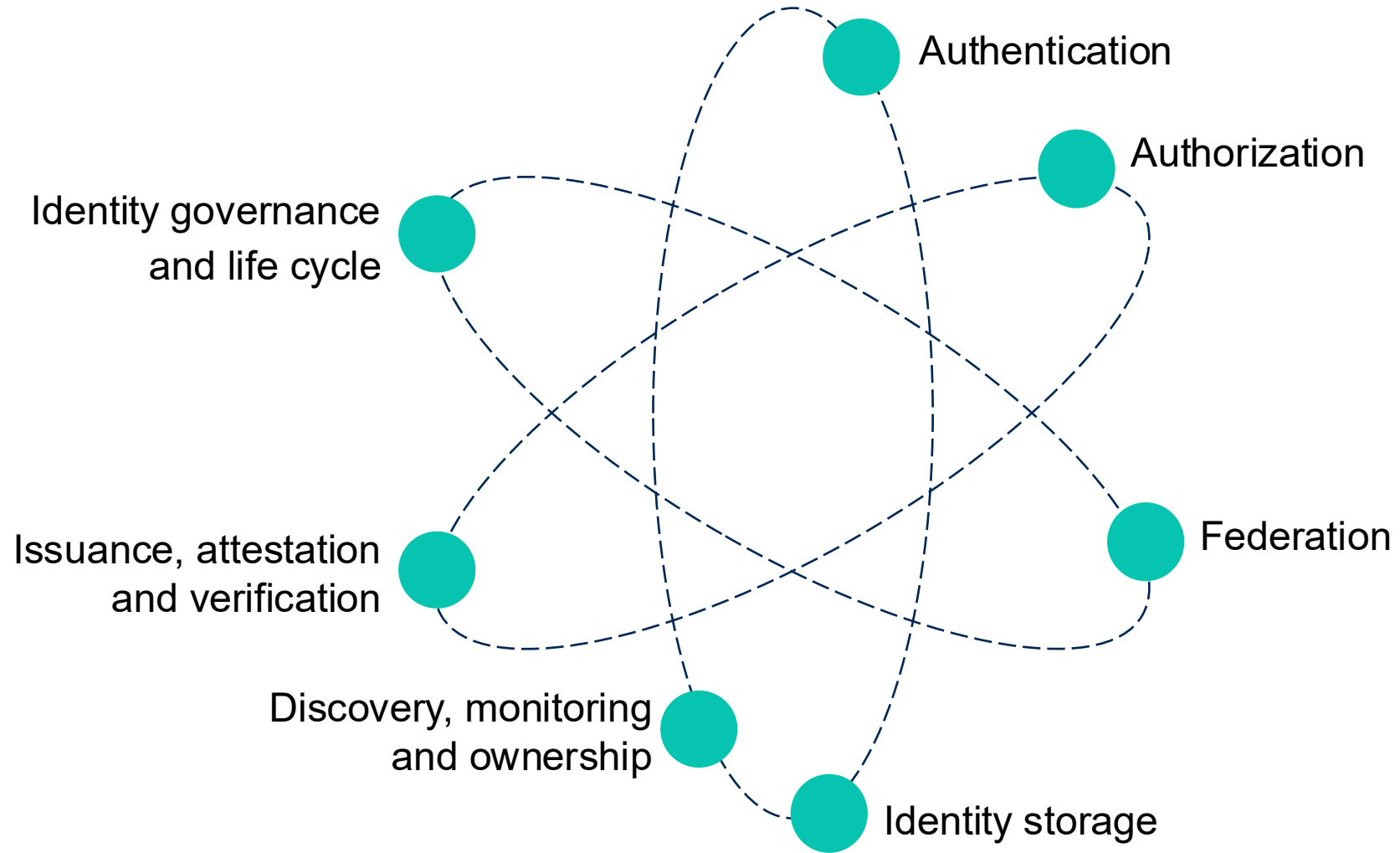
Applications, services and their relationships to infrastructure



# Shift Focus to Workloads Not Credentials



# Identify Missing IAM Controls



# Broaden Discovery

|                              |           |                     |           |
|------------------------------|-----------|---------------------|-----------|
| Easy                         | IAM tools | DevOps environments | IaaS/PaaS |
| Growing list of plugins      | Manual    | SaaS                | Network   |
| Growing list of integrations | End-point |                     |           |

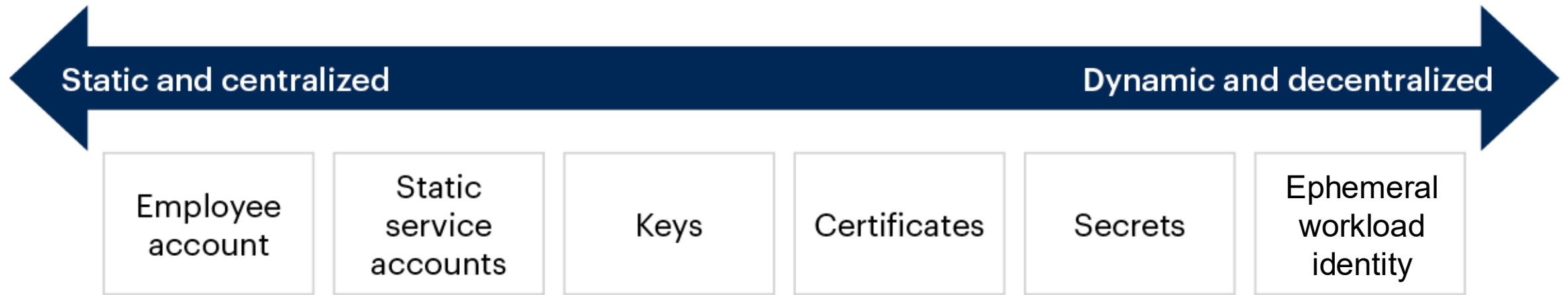
## **Key Issue Take-Away:**

Use the structured IAM taxonomy to clearly define scope, identify blind spots, understand functional requirements and set the right expectations within the organization to improve business outcomes.

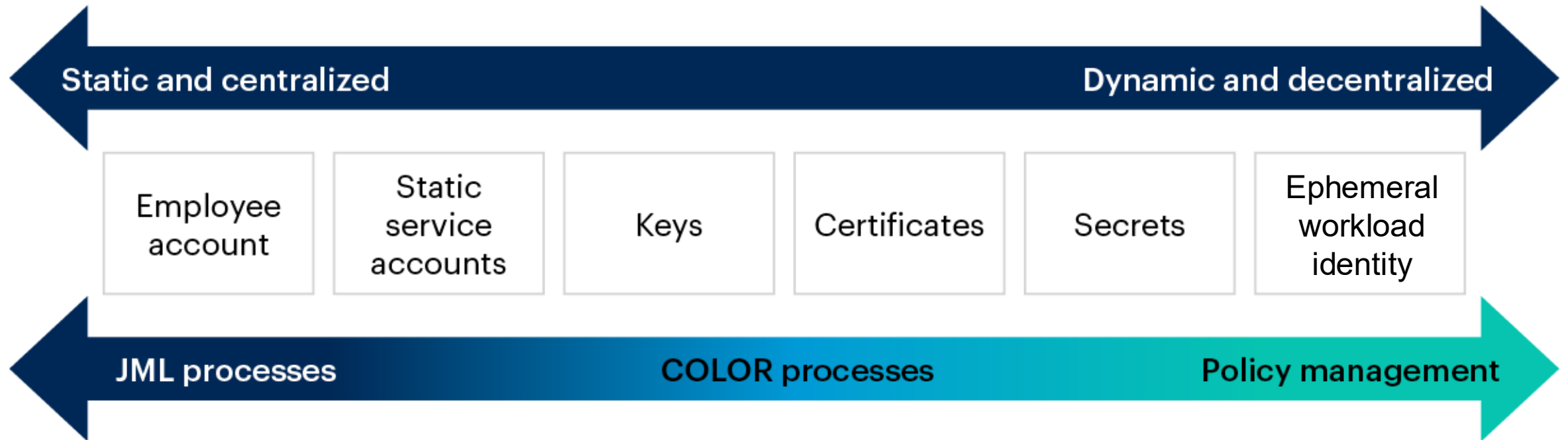
# Key Issues

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# Existing Processes Are Not Helping

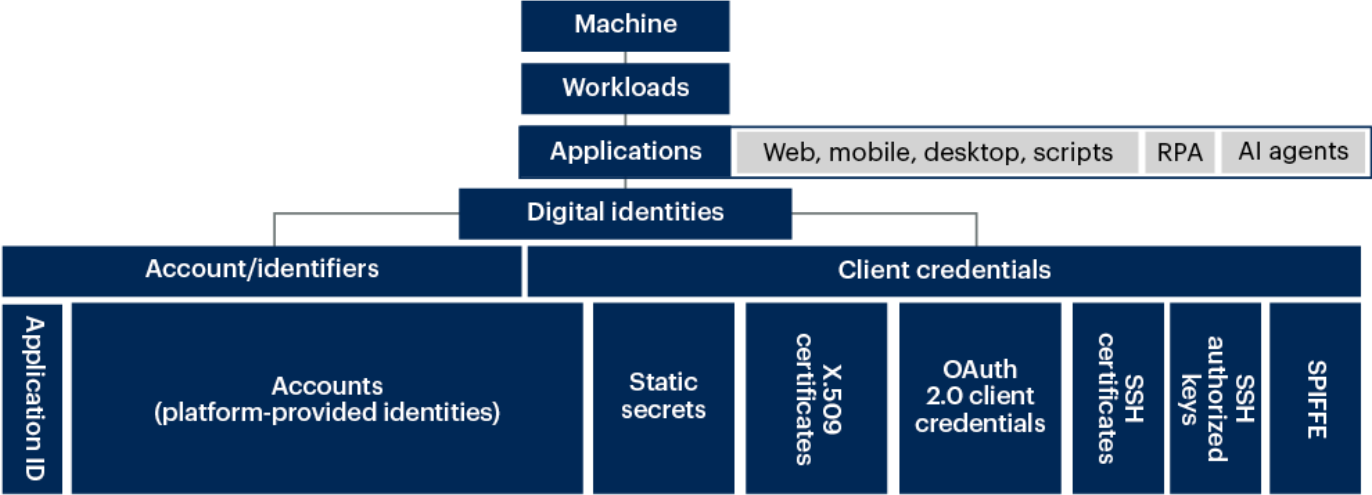


# Tailor Your Processes for the Different Identities



Note: COLOR = Creation, Observability, Life Cycle, Ownership, Remediation; JML = joiners-movers-leavers

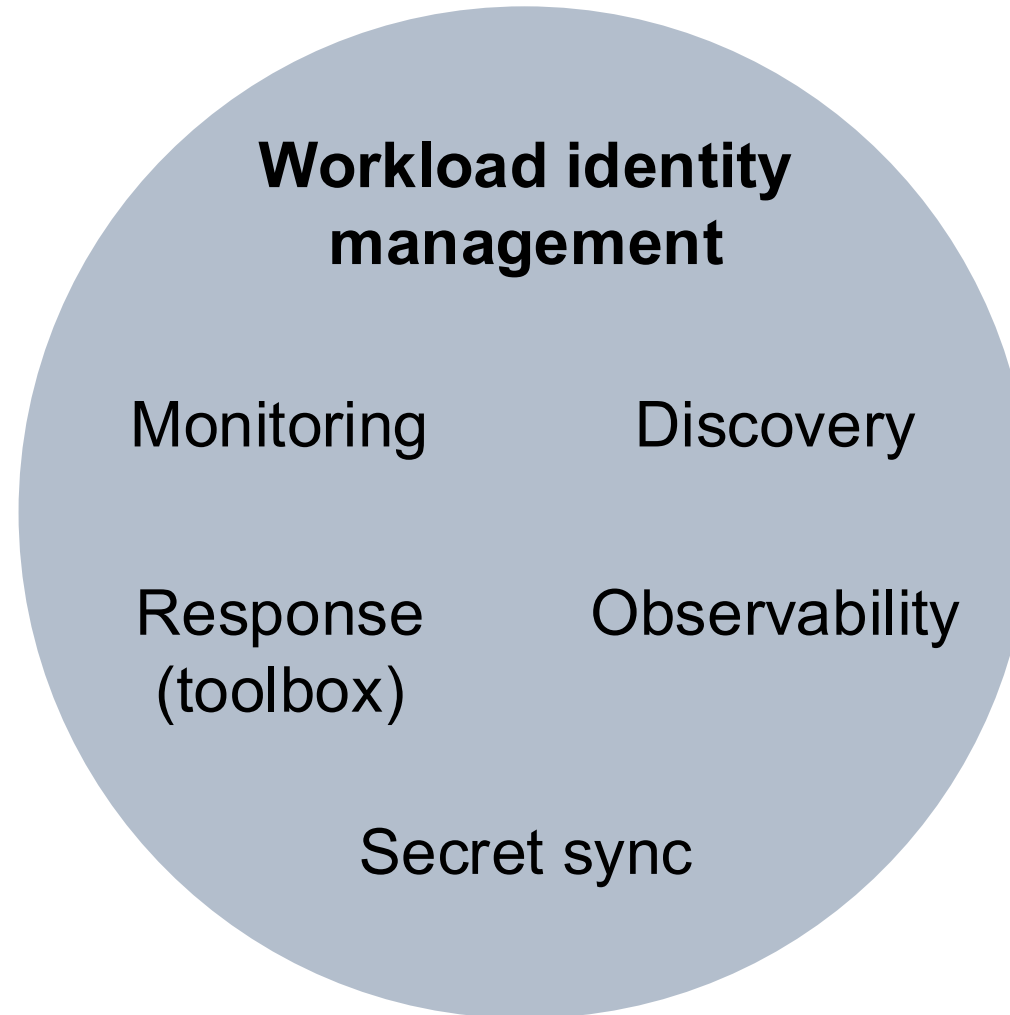
# Identify Your Blind Spots at Level 5 in the Taxonomy



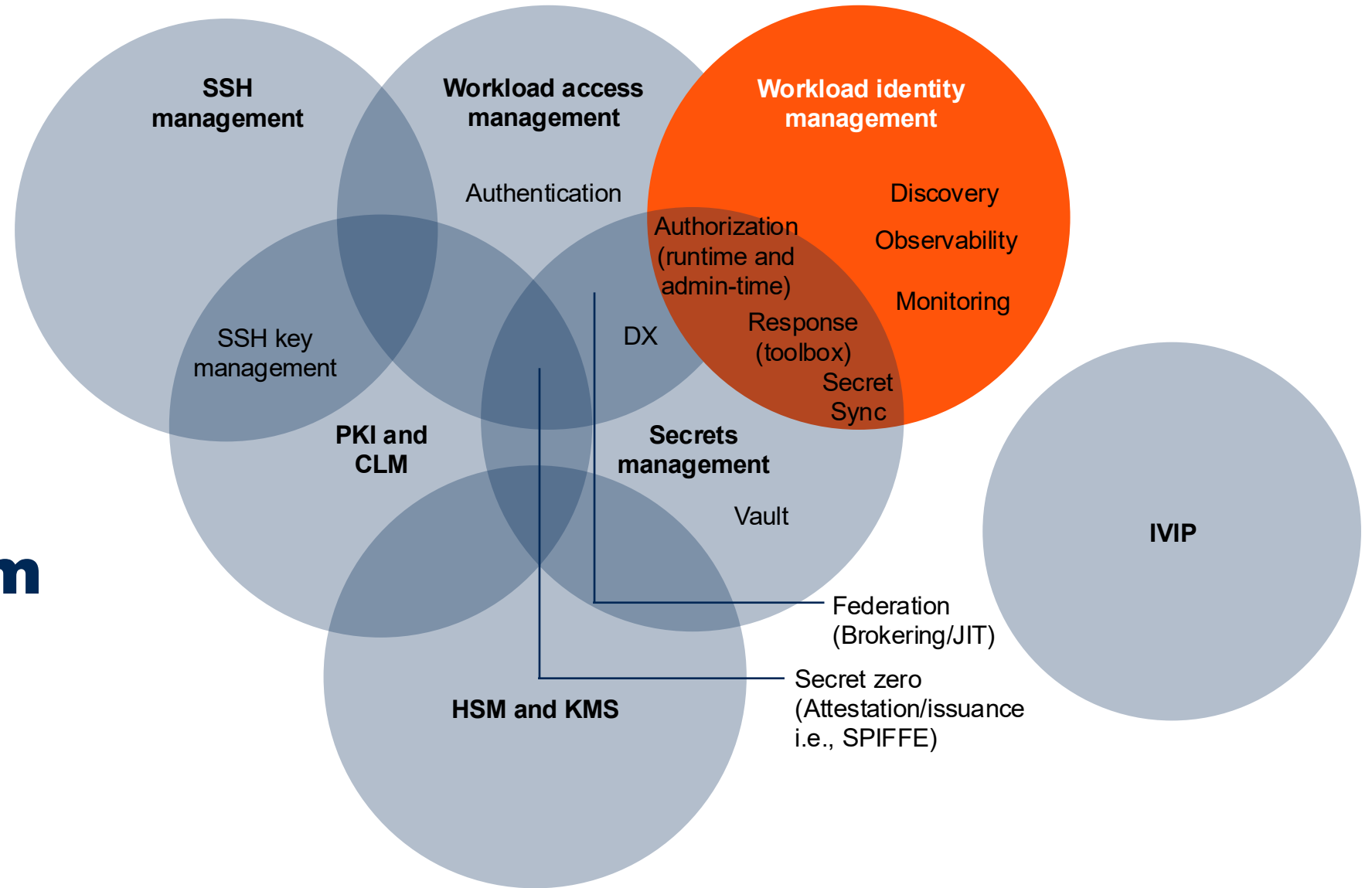
- F** Federation
- Z** Authorization
- A** Authentication
- M** Monitoring and ownership
- G** Identity governance and life cycle
- O** Discovery, monitoring and ownership
- I** Issuance, attestation and verification
- S** Identity storage

Identity fabric

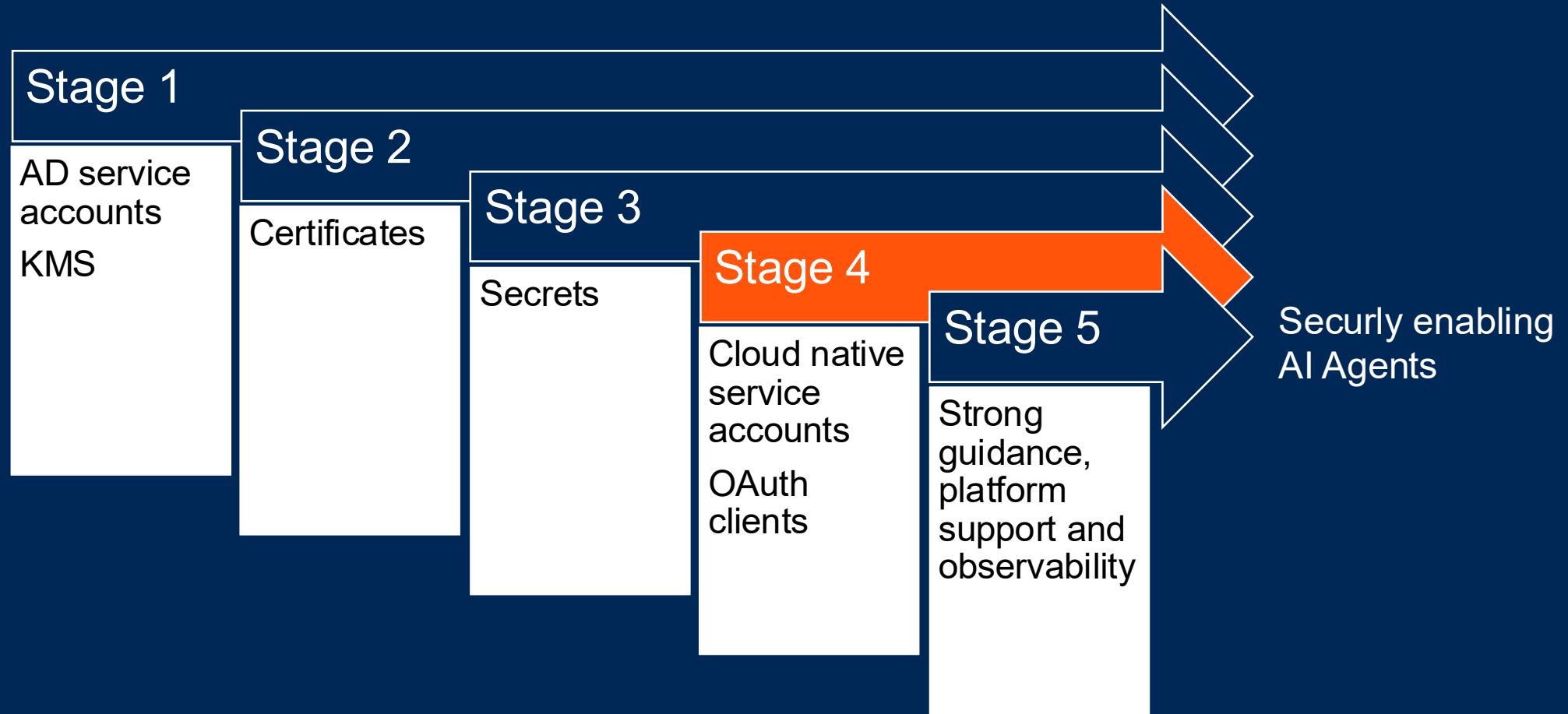
# Workload Identity Management Controls



# Workload IAM — Capability Euler Diagram



# Now, but Take a Phased Approach!



# A Plethora of Example Vendors

## PKI and certificate management

- AppViewX
- Amazon Web Services (AWS)
- CyberArk Software (Venafi)
- DigiCert
- Entrust
- EverTrust
- GlobalSign
- Google Cloud Platform
- Keyfactor
- Microsoft
- IN Groupe
- PKI Solutions
- Sectigo

## Secrets managers

- Akeyless
- BeyondTrust
- CyberArk Software
- Delinea
- Doppler
- Fortanix
- HashiCorp
- Infisical
- OpenBao

## Workload identity management

- Astrix Security
- Clutch
- CrowdStrike
- Entro Security
- GitGuardian
- Hush Security
- Natoma
- Oasis Security
- Token Security
- Veza

## SSH key management

- AppViewX
- BeyondTrust
- CyberArk Software
- Keyfactor
- SSH Communications Security

## KMS and HSM

- Amazon Web Services
- Crypto4A
- Entrust
- Fortanix
- Google Cloud Platform
- Microsoft Azure
- PKI Solutions
- PKWARE
- Thales Group
- Utimaco Management

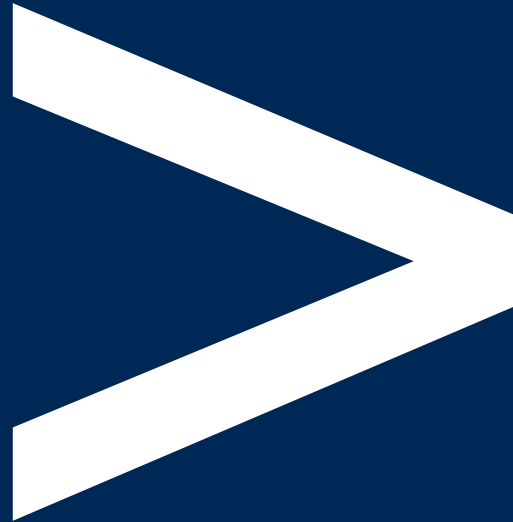
## Workload access management

- Aembit
- Defakto
- Teleport
- TrustFour
- CyberArk Software



# **When Is it Time to Adopt?**

**Complexity**



**Size**

## Key Issue Take-Away:

Identify the right type of tool, assess your current portfolio and your organization's maturity, and add compensating controls using best-of-breed tools to address technical debt from legacy workload identities practices.



# What's next?

- Identity data
- Threat detection
- Posture management
- Workload access management

# Recommendations

- ④ Use an IAM taxonomy to clearly define scope, identify blind spots, understand functional requirements and set the right expectations within the organization to improve business outcomes.
- ④ Create a machine IAM working group.
- ④ Begin by first understanding your current toolsets that contain workload IAM capabilities and then integrating new tools and/or capabilities into your overall strategy.
- ④ Don't underestimate how complex and challenging it is to implement discovery and rotation at scale.
- ④ Establish a continuous process of improvement.

# Recommended Gartner Research

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## 🔍 [IAM — Taxonomy, Domains and Tooling](#)

Erik Wahlstrom

## 🔍 [2026 Planning Guide for Identity and Access Management](#)

Erik Wahlstrom, Mary Ruddy and Others

## 🔍 [Evolve IAM Using 10 Identity Fabric Principles](#)

Erik Wahlstrom and Mary Ruddy

## 🔍 [Strategic Roadmap for Modern Machine IAM](#)

Felix Gaehtgens, Erik Wahlstrom and Steve Wessels